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8 **(a)** **(i)** State what is meant by an electric field.

.....

..... [1]

(ii) Fig. 8.1 shows a negative point charge.

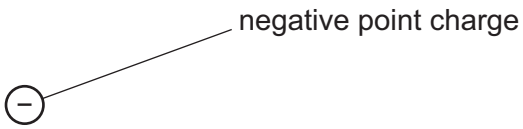

Fig. 8.1

On Fig. 8.1, draw **four** field lines to show the pattern and the direction of the electric field due to the negative point charge. [2]

(b) The potential difference (p.d.) of a lightning strike is 2.9×10^8 V. The energy transferred by the lightning strike is 4.5×10^5 MJ.

Calculate the charge that flows.

charge = ... [3]

(c) The current in a lamp is 2.0 A. The lamp is switched on for a time of 120 s.

Calculate the charge that flows.

charge = ... [2]

[Total: 8]